



TULSIRAMJI GAIKWAD-PATIL College of Engineering and Technology

Wardha Road, Nagpur - 441108

Accredited with NAAC A+ Grade

Approved by AICTE, New Delhi, Govt. of Maharashtra



(An Autonomous Institute Affiliated to RTM Nagpur University with NAAC A+ Grade)

Department of Information Technology

Session 2024-25(Even Semester)

VISION	MISSION
To emerge as a learning hub and center of excellence in the domain of Information Technology	[M1] To impart quality technical education through effective teaching learning process. [M2] To provide a platform to address societal issues as well as challenges faced by IT industries. [M3] To foster a culture of research and impart innovative and entrepreneurial skills in the field of IT. [M4] To ensure overall development of students and staff by inculcating knowledge and professional ethics as a part of lifelong learning.

END SEMESTER

Semester: IV

Programme: B. Tech Information Technology

Course: Database Management System

Time: 3 Hours

Max Marks: 60 Marks

Instructions to candidates:

1. All questions are compulsory in Group A
2. Solve any two sub-question out of three in Group B
3. Assume suitable data wherever necessary
4. All question carries marks as indicated

BIT32402	Course Outcome
CO1	Understand the fundamental of database system and DBMS concepts.
CO2	Identify the concept of conceptual data modeling using the E/R model.
CO3	Illustrate the concept of embedded SQL and its role in application development.
CO4	Analyze the process of database normalization and the steps for achieving 1NF,2NF,3NF,and BCNF
CO5	Compare Various indexing techniques like hash-based indexing,dynamic hashing,and B+trees.

Group A				
Que. 1	Answer the following Question	CO	BTL	Marks
a.	What is data model?	CO1	2	2
b.	Define primary key.	CO2	2	2
c.	What is DDL?	CO3	2	2
d.	Define functional dependency.	CO4	3	2
e.	What is indexing?	CO5	2	2
Group B				
Que. 2	Answer the following Question (Solve any Two)	CO	BTL	Marks

a.	Explain database users and their roles.	CO1	2	5
b.	Describe data abstraction in DBMS.	CO1	2	5
c.	Explain challenges in DBMS.	CO1	2	5
Que 3.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain attributes and their types.	CO2	2	5
b.	Describe different types of keys	CO2	3	5
c.	Explain relational calculus.	CO2	3	5
Que 4.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain SELECT-FROM-WHERE clause with example.	CO3	2	5
b.	Describe aggregate functions in SQL.	CO3	3	5
c.	Explain correlated and uncorrelated queries.	CO3	4	5
Que 5.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain Armstrong's axioms.	CO4	2	5
b.	Describe decomposition in normalization.	CO4	3	5
c.	Explain multivalued dependency and 4NF.	CO4	4	5
Que 6.	Answer the following Question (Solve any Two)	CO	BTL	Marks
a.	Explain hashing techniques.	CO5	2	5
b.	Describe file organization methods.	CO5	3	5
c.	Explain recovery techniques (undo/redo).	CO5	4	5

Bloom's Taxonomy Level (BTL)- Remember , Understand , Apply, Analyze , Evaluate , Create